Kernix 0.12.0 / Performance brief

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

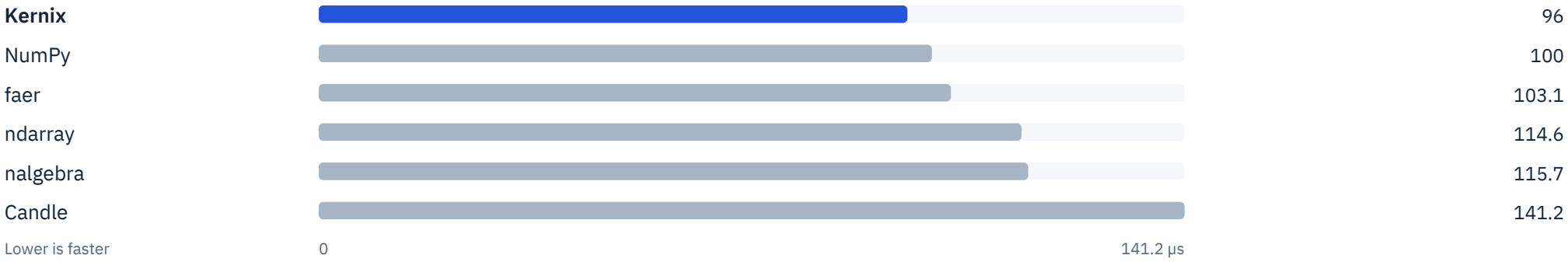
51 workloads

170 observations

Source precision retained

Matrix multiplication

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · self product



How to read this report

Allocation-inclusive API latency unless marked reused output. Peer medians are observations.

Charts round to 6 significant digits (≈). The exact-value appendix preserves all recorded digits.

Kernix 0.12.0 / Sparse matrices & memory

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Sparse matrix × vector

Per-call median

2048 × 2048 · five diagonals · f64 · 1 worker · five diagonals · prepared operands · owned output · finite contracts differ



Sparse × dense matrix

Per-call median

512 × 512 · 32 RHS · five diagonals · f64 · 1 worker · five diagonals · prepared operands · owned output · finite contracts differ



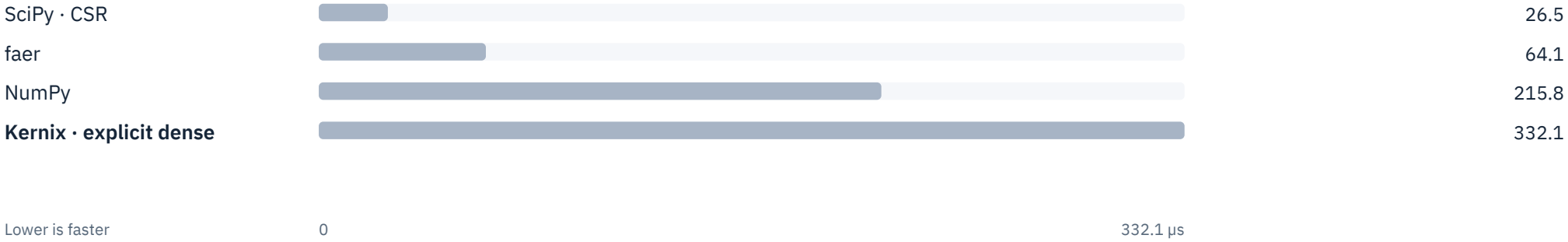
Kernix 0.12.0 / Sparse matrices & memory

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Sparse × dense matrix

Per-call median

512 × 512 · 32 RHS · five diagonals · f32 · 1 worker · five diagonals · prepared operands · owned output · finite contracts differ



Sparse × sparse matrix

Per-call median

256 × 256 · five diagonals each · f64 · 1 worker · five diagonals · prepared operands · owned output · finite contracts differ



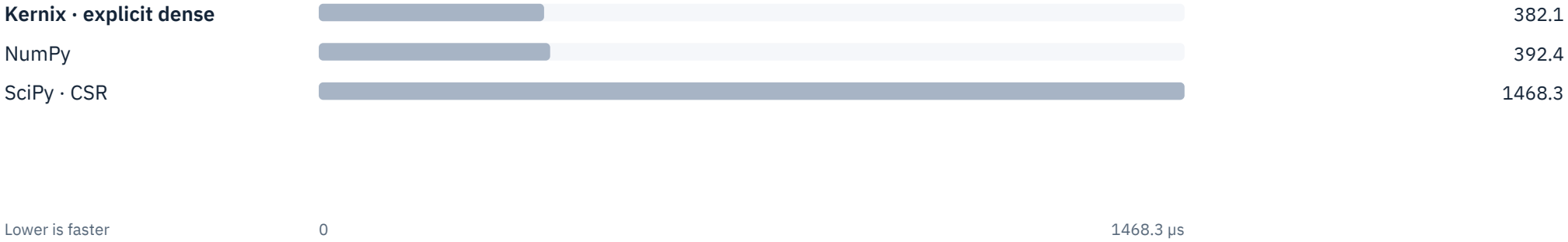
Kernix 0.12.0 / Sparse matrices & memory

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Automatic CSR construction

Per-call median

512 × 512 · clone + scan + compress · f64 · Construction · dense copy or compression · preparation costs included



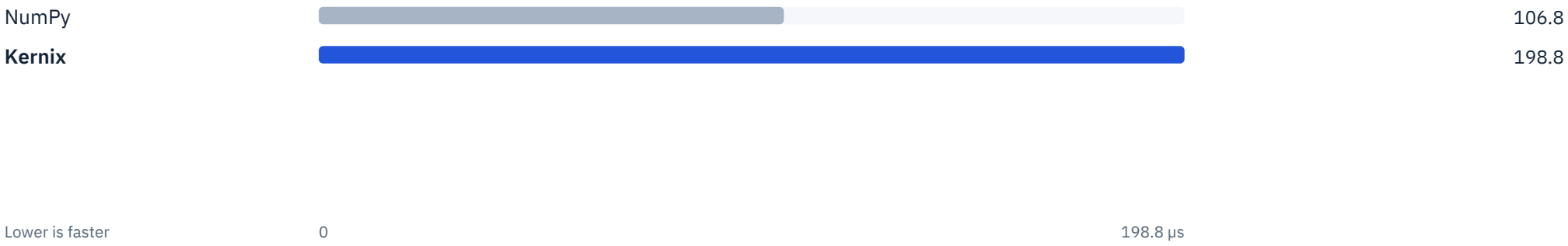
Kernix 0.12.0 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication · transposed view

Per-call median

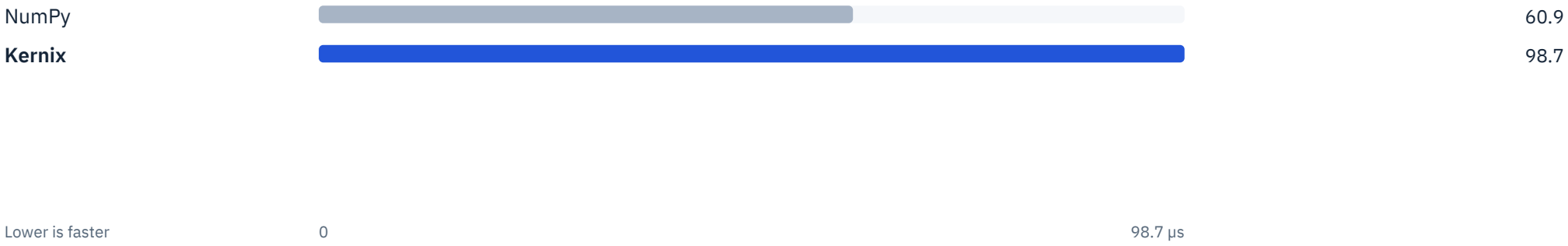
128 × 128 · 128 × 128 · f64 · 1 worker · borrowed transpose · allocated output · independent operands



Matrix multiplication · transposed view

Per-call median

128 × 128 · 128 × 128 · f32 · 1 worker · borrowed transpose · allocated output · independent operands



Kernix 0.12.0 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Independent matrix operands

Per-call median

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · independent operands



Independent matrix operands

Per-call median

128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · independent operands



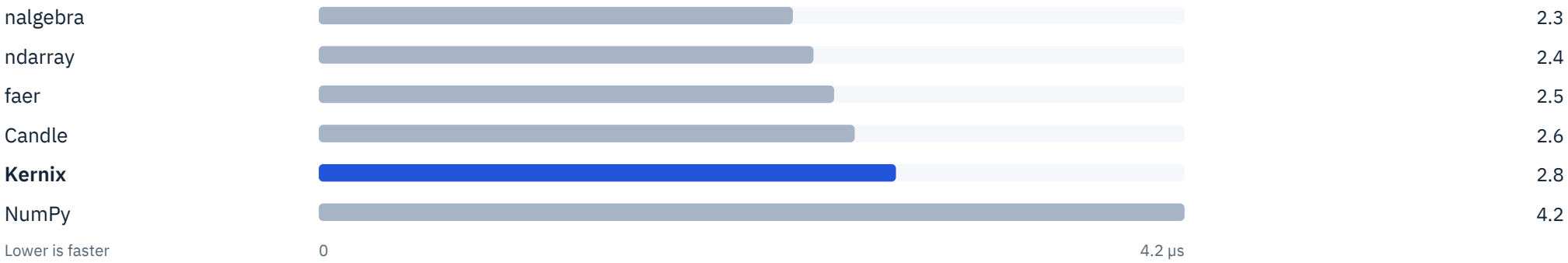
Kernix 0.12.0 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

32 × 32 · 32 × 32 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f64 · 1 worker · allocated output · self product



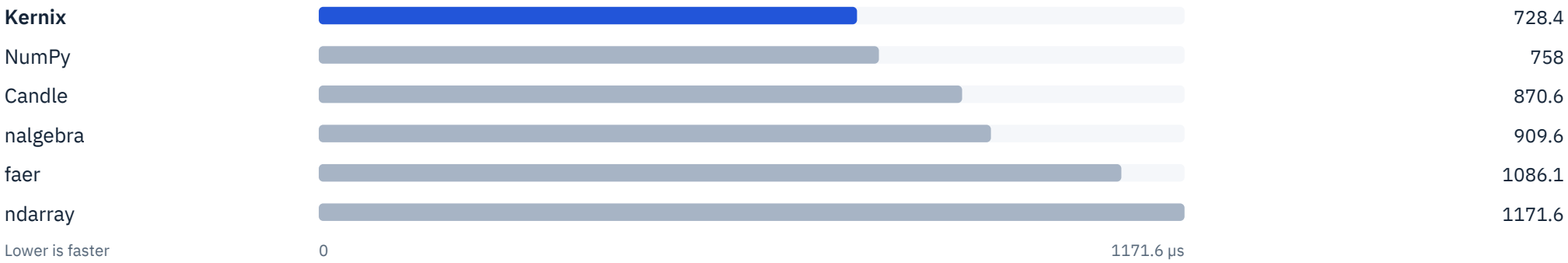
Kernix 0.12.0 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

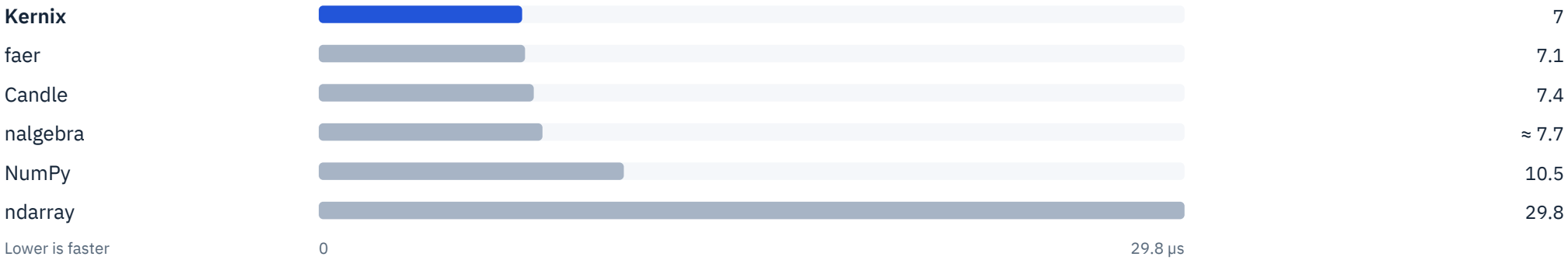
256 × 256 · 256 × 256 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f32 · 1 worker · allocated output · self product



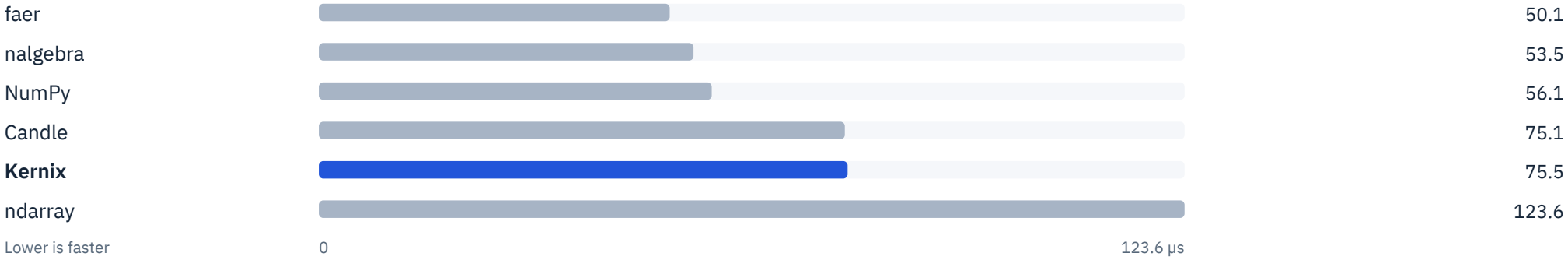
Kernix 0.12.0 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

256 × 256 · 256 × 256 · f32 · 1 worker · allocated output · self product



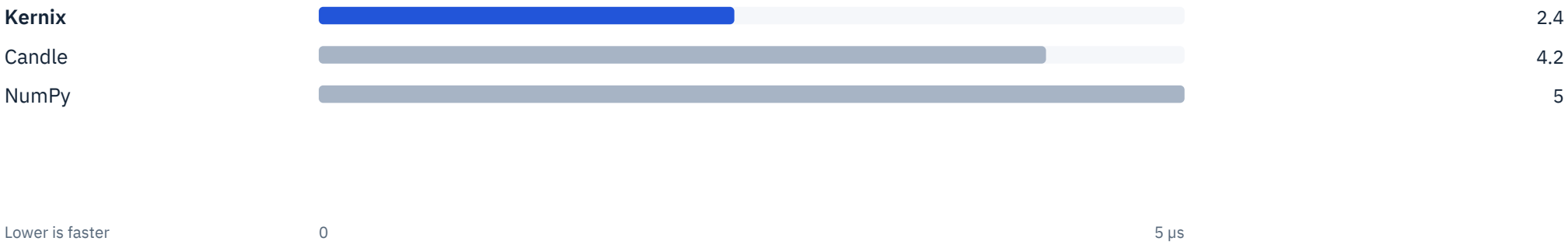
Kernix 0.12.0 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Batched multiplication

Per-call median

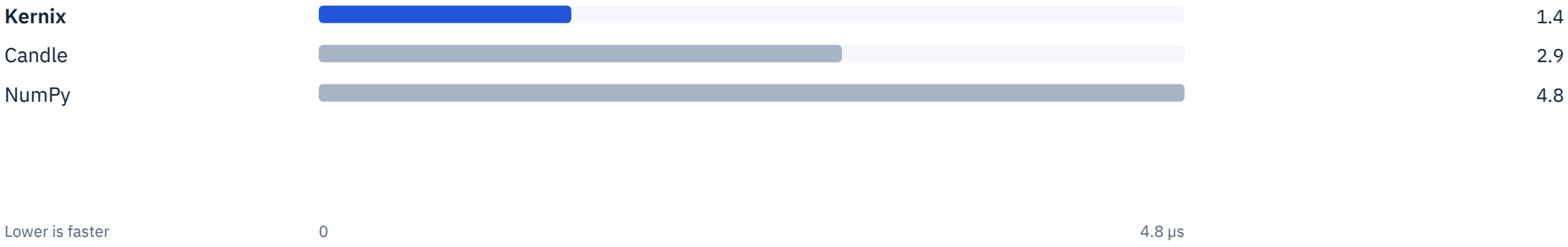
4 × 16 × 32 · 1 × 32 × 16 · f64 · 1 worker · allocated output · broadcast batches



Batched multiplication

Per-call median

4 × 16 × 32 · 1 × 32 × 16 · f32 · 1 worker · allocated output · broadcast batches



Kernix 0.12.0 / Explicit parallel execution

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Matmul · 2 workers · reused output

Per-call median

64 × 64 · 64 × 64 · f64 · 2 workers · reused output · pool preparation excluded



Matmul · 2 workers · reused output

Per-call median

128 × 128 · 128 × 128 · f64 · 2 workers · reused output · pool preparation excluded



Kernix 0.12.0 / Explicit parallel execution

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Matmul · 2 workers · reused output

Per-call median

256 × 256 · 256 × 256 · f64 · 2 workers · reused output · pool preparation excluded



Matmul · 2 workers · reused output

Per-call median

128 × 128 · 128 × 128 · f32 · 2 workers · reused output · pool preparation excluded



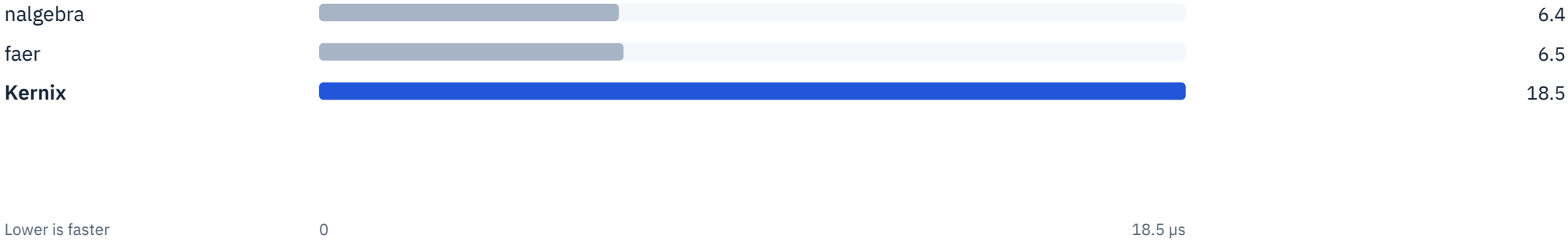
Kernix 0.12.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky solve · reused factor

Per-call median

128 × 128, 1 RHS · f64 · 1 worker · prepared reusable factor · owned output · factorization excluded



Cholesky solve · reused factor

Per-call median

256 × 256, 64 RHS · f64 · 1 worker · prepared reusable factor · owned output · factorization excluded



Kernix 0.12.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky solve · reused factor

Per-call median

512 × 512, 128 RHS · f64 · 1 worker · prepared reusable factor · owned output · factorization excluded



LU solve · reused factor

Per-call median

256 × 256, 64 RHS · f64 · 1 worker · prepared reusable factor · owned output · factorization excluded



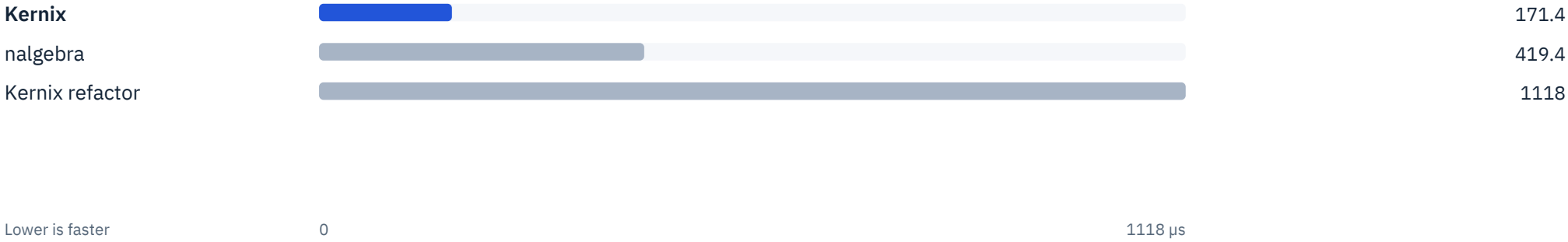
Kernix 0.12.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky rank-one update

Per-call median

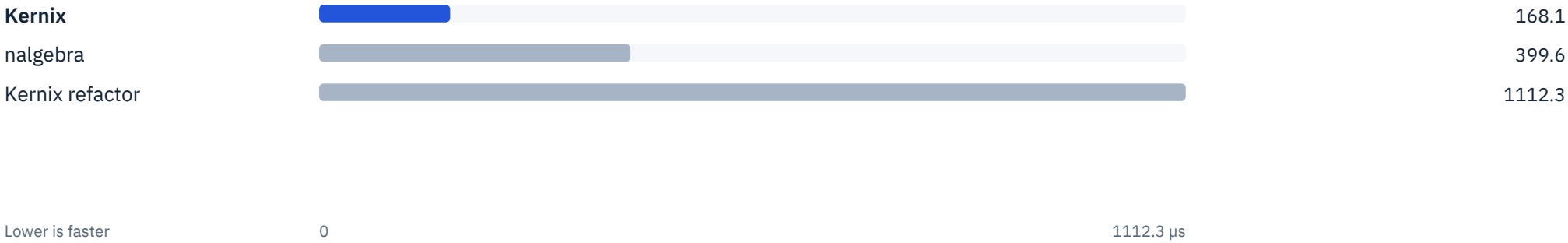
256 × 256 · f64 · Prepared factors · owned output · refactor uses preformed modified matrix



Cholesky rank-one downdate

Per-call median

256 × 256 · f64 · Prepared factors · owned output · refactor uses preformed modified matrix



Kernix 0.12.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

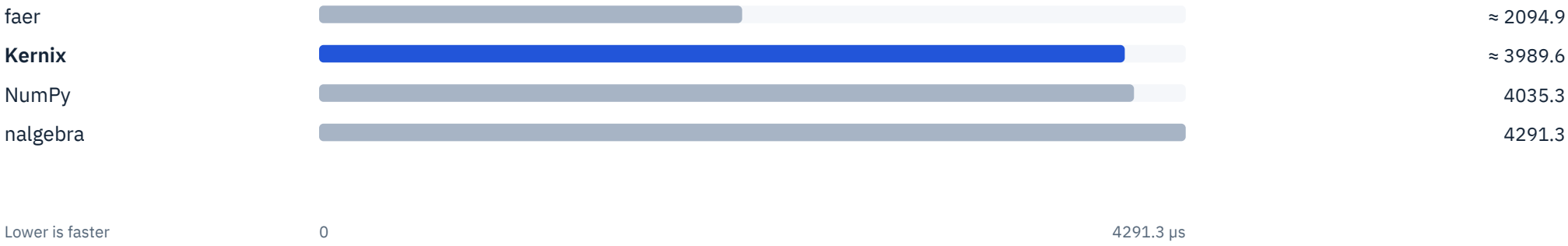
Cholesky rank-one update

512 × 512 · f64 · Prepared factors · owned output · refactor uses preformed modified matrix



Thin QR

512 × 128 · f64 · 1 worker · allocated output

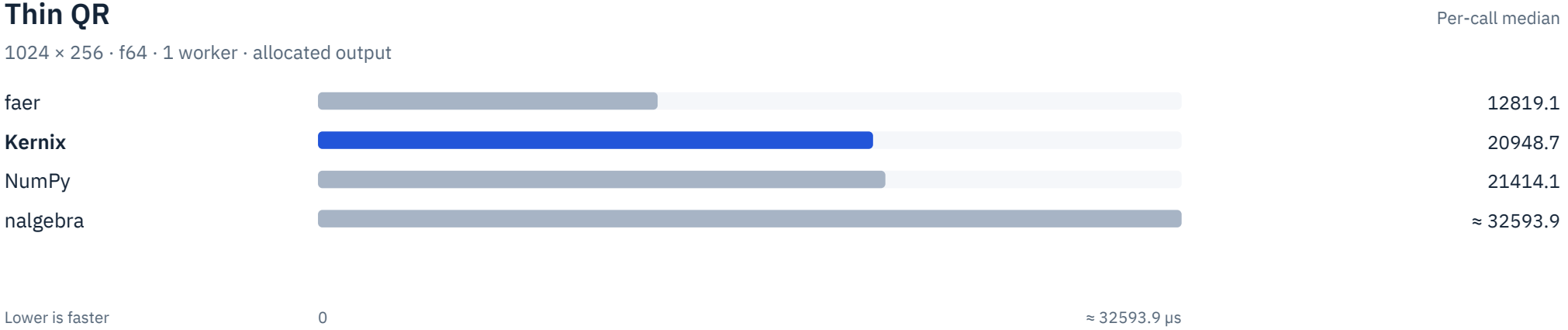


Kernix 0.12.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

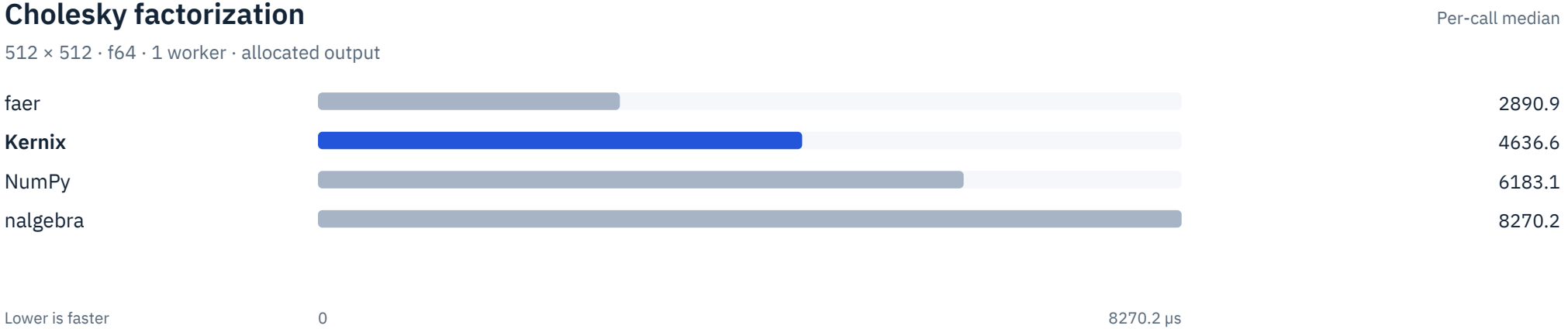
Thin QR

1024 × 256 · f64 · 1 worker · allocated output



Cholesky factorization

512 × 512 · f64 · 1 worker · allocated output



Kernix 0.12.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Column-pivoted QR

96 × 32 · f64 · 1 worker · allocated output

Per-call median



Column-pivoted QR

256 × 64 · f64 · 1 worker · allocated output

Per-call median



Kernix 0.12.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Minimum-norm least squares

Per-call median

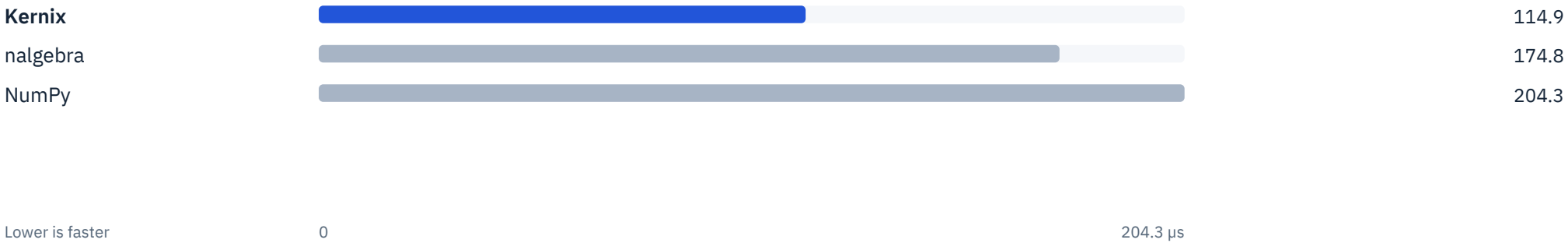
64 × 32, rank 16, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ



Minimum-norm least squares

Per-call median

32 × 64, rank 32, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ

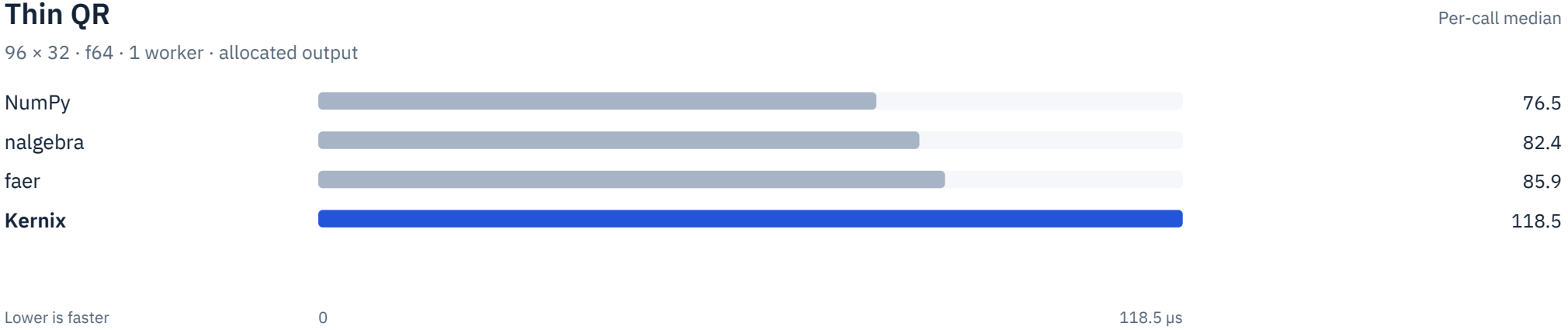


Kernix 0.12.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Thin QR

96 × 32 · f64 · 1 worker · allocated output



Thin QR

256 × 64 · f64 · 1 worker · allocated output



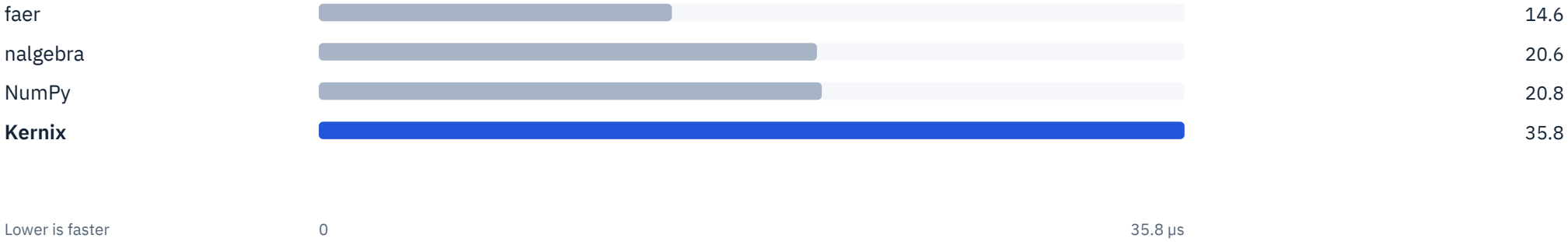
Kernix 0.12.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

Per-call median

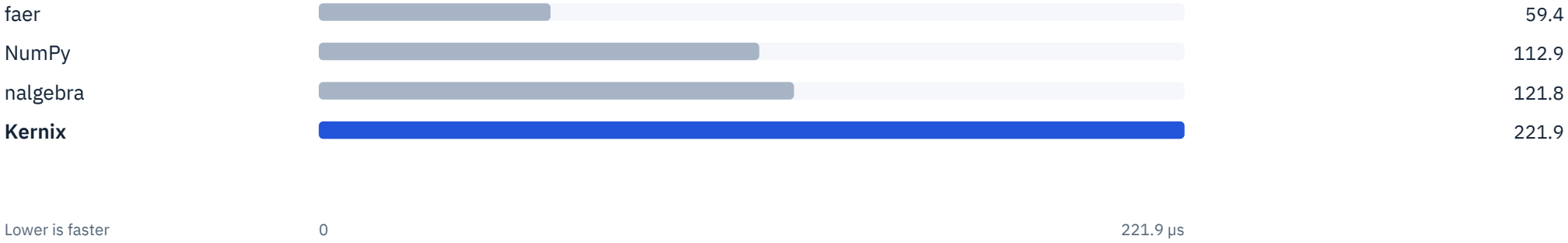
64 × 64 · f64 · 1 worker · allocated output



Cholesky factorization

Per-call median

128 × 128 · f64 · 1 worker · allocated output



Kernix 0.12.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

256 × 256 · f64 · 1 worker · allocated output

Per-call median



Kernix 0.12.0 / Training & autodiff

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Shared-variable backward

Per-call median

sum(X * X), 64 × 32; prepared graph · f64 · 1 worker · prepared graph · fresh gradients



Record + backward

Per-call median

sum(A · W), 32 × 16 · 16 × 8 · f64 · 1 worker · allocated output · self product

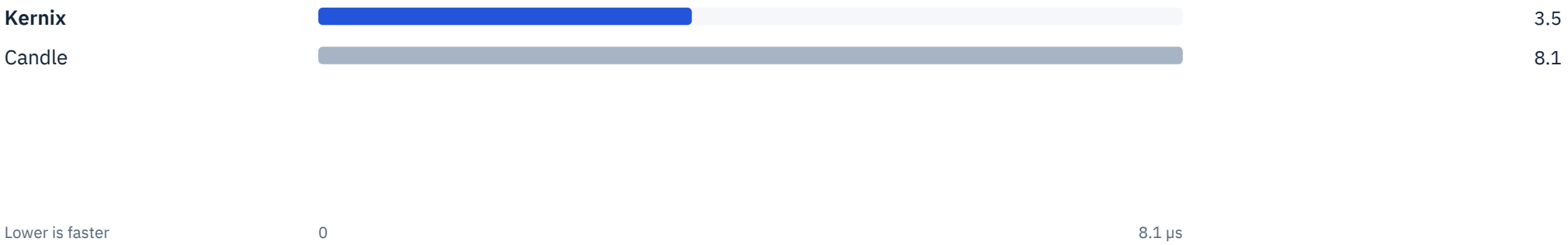


Kernix 0.12.0 / Training & autodiff

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Model inference

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Model recording + forward

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Kernix 0.12.0 / Training & autodiff

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Model backward

Per-call median

prepared graph → fresh gradients · f64 · 1 worker · prepared graph · fresh gradients



Parameter update

Per-call median

transactional SGD, fixed gradients · f64 · 1 worker · preparation as documented

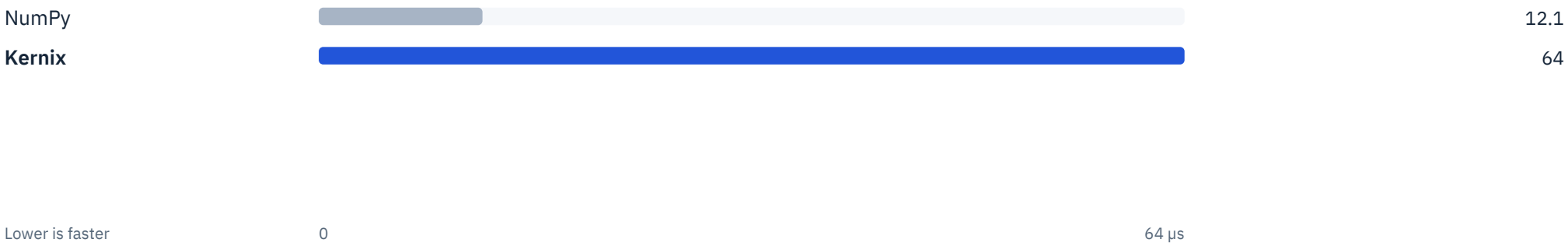


Kernix 0.12.0 / Tensor contractions

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Coalesced tensor contraction

16 × 32 × 8 · 32 × 8 × 16 · f64 · 1 worker · allocated output

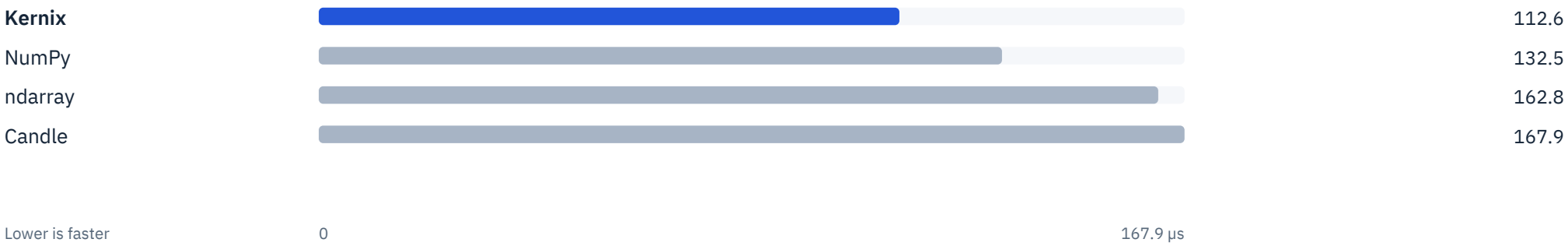


Kernix 0.12.0 / Statistics & activations

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Row softmax

256 × 64 · f64 · 1 worker · allocated output



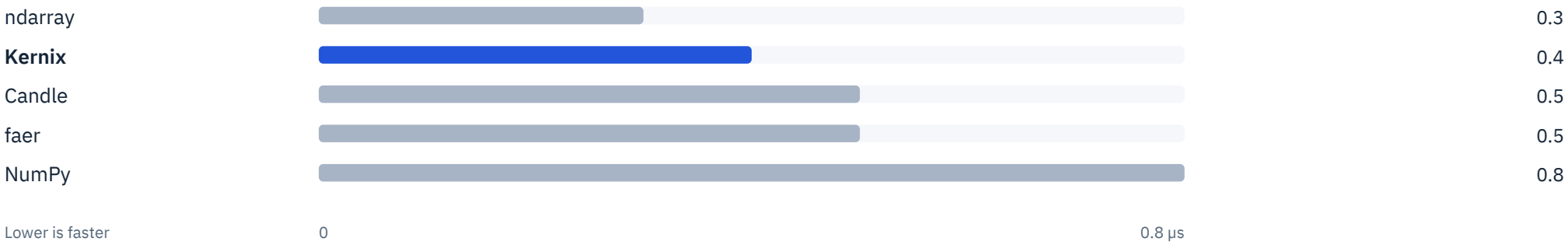
Kernix 0.12.0 / Tensor construction

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Create zeros

Per-call median

64 × 64 · f32 · 1 worker · allocated output



Create ones

Per-call median

64 × 64 · f32 · 1 worker · allocated output



Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Sparse matrix × vector / SciPy · CSR 2048 × 2048 · five diagonals · f64	11300	11.3
Sparse matrix × vector / NumPy 2048 × 2048 · five diagonals · f64	655900	655.9
Sparse matrix × vector / faer 2048 × 2048 · five diagonals · f64	7900.000000000001	7.900000000000001
Sparse matrix × vector / Kernix · explicit dense 2048 × 2048 · five diagonals · f64	2669800	2669.8
Sparse × dense matrix / SciPy · CSR 512 × 512 · 32 RHS · five diagonals · f64	41600	41.6
Sparse × dense matrix / Kernix · explicit dense 512 × 512 · 32 RHS · five diagonals · f64	626300	626.3
Sparse × dense matrix / NumPy 512 × 512 · 32 RHS · five diagonals · f64	466300	466.3
Sparse × dense matrix / faer 512 × 512 · 32 RHS · five diagonals · f64	67700	67.7
Sparse × dense matrix / SciPy · CSR 512 × 512 · 32 RHS · five diagonals · f32	26500	26.5
Sparse × dense matrix / Kernix · explicit dense 512 × 512 · 32 RHS · five diagonals · f32	332100	332.1

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Sparse × dense matrix / faer 512 × 512 · 32 RHS · five diagonals · f32	64100	64.1
Sparse × dense matrix / NumPy 512 × 512 · 32 RHS · five diagonals · f32	215800	215.8
Sparse × sparse matrix / SciPy · CSR 256 × 256 · five diagonals each · f64	66800	66.8
Sparse × sparse matrix / NumPy 256 × 256 · five diagonals each · f64	760000	760
Sparse × sparse matrix / Kernix · explicit dense 256 × 256 · five diagonals each · f64	745700	745.7
Automatic CSR construction / NumPy 512 × 512 · clone + scan + compress · f64	392400	392.4
Automatic CSR construction / SciPy · CSR 512 × 512 · clone + scan + compress · f64	1468300	1468.3
Automatic CSR construction / Kernix · explicit dense 512 × 512 · clone + scan + compress · f64	382100	382.1
Matrix multiplication · transposed view / NumPy 128 × 128 · 128 × 128 · f64	106800	106.8
Matrix multiplication · transposed view / Kernix 128 × 128 · 128 × 128 · f64	198800	198.8

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication · transposed view / Kernix 128 × 128 · 128 × 128 · f32	98700	98.7
Matrix multiplication · transposed view / NumPy 128 × 128 · 128 × 128 · f32	60900	60.9
Independent matrix operands / NumPy 128 × 128 · 128 × 128 · f64	101200	101.2
Independent matrix operands / Kernix 128 × 128 · 128 × 128 · f64	141800	141.8
Independent matrix operands / Kernix 128 × 128 · 128 × 128 · f32	75500	75.5
Independent matrix operands / NumPy 128 × 128 · 128 × 128 · f32	56400	56.4
Matrix multiplication / faer 32 × 32 · 32 × 32 · f64	2500	2.5
Matrix multiplication / nalgebra 32 × 32 · 32 × 32 · f64	2300	2.3
Matrix multiplication / NumPy 32 × 32 · 32 × 32 · f64	4200	4.2
Matrix multiplication / ndarray 32 × 32 · 32 × 32 · f64	2400	2.4

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Candle 32 × 32 · 32 × 32 · f64	2600	2.6
Matrix multiplication / Kernix 32 × 32 · 32 × 32 · f64	2800	2.8
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f64	14200	14.2
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f64	57800	57.8
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f64	16000	16
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f64	13200	13.2
Matrix multiplication / faer 64 × 64 · 64 × 64 · f64	14500	14.5
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f64	58200	58.2
Matrix multiplication / faer 128 × 128 · 128 × 128 · f64	103100	103.1
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f64	114600	114.6

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f64	100000	100
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f64	115700	115.7
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f64	96000	96
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f64	141200	141.2
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f64	758000	758
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f64	870600	870.6
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f64	909600	909.6
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f64	728400	728.4
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f64	1171600	1171.6
Matrix multiplication / faer 256 × 256 · 256 × 256 · f64	1086100	1086.1

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f32	7000	7
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f32	7400	7.4
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f32	10500	10.5
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f32	7700.000000000001	7.700000000000001
Matrix multiplication / faer 64 × 64 · 64 × 64 · f32	7100	7.1
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f32	29800	29.8
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f32	75500	75.5
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f32	75100	75.1
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f32	56100	56.1
Matrix multiplication / faer 128 × 128 · 128 × 128 · f32	50100	50.1

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f32	53500	53.5
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f32	123600	123.6
Matrix multiplication / faer 256 × 256 · 256 × 256 · f32	409800	409.8
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f32	427300	427.3
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f32	389700	389.7
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f32	395000	395
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f32	387100	387.1
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f32	427600	427.6
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f64	5000	5
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f64	4200	4.2

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f64	2400	2.4
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f32	2900	2.9
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f32	1400	1.4
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f32	4800	4.8
Matmul · 2 workers · reused output / Kernix 64 × 64 · 64 × 64 · f64	56100	56.1
Matmul · 2 workers · reused output / Kernix 128 × 128 · 128 × 128 · f64	82600	82.6
Matmul · 2 workers · reused output / Kernix 256 × 256 · 256 × 256 · f64	441100	441.1
Matmul · 2 workers · reused output / Kernix 128 × 128 · 128 × 128 · f32	56900	56.9
Cholesky solve · reused factor / nalgebra 128 × 128, 1 RHS · f64	6400	6.4
Cholesky solve · reused factor / Kernix 128 × 128, 1 RHS · f64	18500	18.5

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky solve · reused factor / faer 128 × 128, 1 RHS · f64	6500	6.5
Cholesky solve · reused factor / faer 256 × 256, 64 RHS · f64	326000	326
Cholesky solve · reused factor / nalgebra 256 × 256, 64 RHS · f64	1320100	1320.1
Cholesky solve · reused factor / Kernix 256 × 256, 64 RHS · f64	1296300	1296.3
Cholesky solve · reused factor / nalgebra 512 × 512, 128 RHS · f64	10219600	10219.6
Cholesky solve · reused factor / faer 512 × 512, 128 RHS · f64	2264300	2264.3
Cholesky solve · reused factor / Kernix 512 × 512, 128 RHS · f64	9565900	9565.9
LU solve · reused factor / nalgebra 256 × 256, 64 RHS · f64	1154800	1154.8
LU solve · reused factor / faer 256 × 256, 64 RHS · f64	372100	372.1
LU solve · reused factor / Kernix 256 × 256, 64 RHS · f64	1306100	1306.1

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky rank-one update / Kernix refactor 256 × 256 · f64	1118000	1118
Cholesky rank-one update / nalgebra 256 × 256 · f64	419400	419.4
Cholesky rank-one update / Kernix 256 × 256 · f64	171400	171.4
Cholesky rank-one downdate / Kernix refactor 256 × 256 · f64	1112300	1112.3
Cholesky rank-one downdate / nalgebra 256 × 256 · f64	399600	399.6
Cholesky rank-one downdate / Kernix 256 × 256 · f64	168100	168.1
Cholesky rank-one update / Kernix refactor 512 × 512 · f64	4671400	4671.4
Cholesky rank-one update / Kernix 512 × 512 · f64	1351300	1351.3
Cholesky rank-one update / nalgebra 512 × 512 · f64	559800	559.8
Thin QR / Kernix 512 × 128 · f64	3989600.0000000005	3989.6000000000005

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Thin QR / NumPy 512 × 128 · f64	4035300	4035.3
Thin QR / faer 512 × 128 · f64	2094899.9999999998	2094.8999999999998
Thin QR / nalgebra 512 × 128 · f64	4291300	4291.3
Thin QR / nalgebra 1024 × 256 · f64	32593900.000000004	32593.900000000004
Thin QR / Kernix 1024 × 256 · f64	20948700	20948.7
Thin QR / faer 1024 × 256 · f64	12819100	12819.1
Thin QR / NumPy 1024 × 256 · f64	21414100	21414.1
Cholesky factorization / faer 512 × 512 · f64	2890900	2890.9
Cholesky factorization / Kernix 512 × 512 · f64	4636600	4636.6
Cholesky factorization / nalgebra 512 × 512 · f64	8270200	8270.2

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky factorization / NumPy 512 × 512 · f64	6183100	6183.1
Column-pivoted QR / faer 96 × 32 · f64	71900	71.9
Column-pivoted QR / Kernix 96 × 32 · f64	125000	125
Column-pivoted QR / nalgebra 96 × 32 · f64	108300	108.3
Column-pivoted QR / Kernix 256 × 64 · f64	925800	925.8
Column-pivoted QR / faer 256 × 64 · f64	413400	413.4
Column-pivoted QR / nalgebra 256 × 64 · f64	906000	906
Minimum-norm least squares / Kernix 64 × 32, rank 16, 4 RHS · f64	96600	96.6
Minimum-norm least squares / NumPy 64 × 32, rank 16, 4 RHS · f64	112700	112.7
Minimum-norm least squares / nalgebra 64 × 32, rank 16, 4 RHS · f64	103500	103.5

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Minimum-norm least squares / NumPy 32 × 64, rank 32, 4 RHS · f64	204300	204.3
Minimum-norm least squares / Kernix 32 × 64, rank 32, 4 RHS · f64	114900	114.9
Minimum-norm least squares / nalgebra 32 × 64, rank 32, 4 RHS · f64	174800	174.8
Thin QR / Kernix 96 × 32 · f64	118500	118.5
Thin QR / faer 96 × 32 · f64	85900	85.9
Thin QR / NumPy 96 × 32 · f64	76500	76.5
Thin QR / nalgebra 96 × 32 · f64	82400	82.4
Thin QR / nalgebra 256 × 64 · f64	650300	650.3
Thin QR / NumPy 256 × 64 · f64	546600	546.6
Thin QR / faer 256 × 64 · f64	386700	386.7

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Thin QR / Kernix 256 × 64 · f64	751000	751
Cholesky factorization / NumPy 64 × 64 · f64	20800	20.8
Cholesky factorization / Kernix 64 × 64 · f64	35800	35.8
Cholesky factorization / nalgebra 64 × 64 · f64	20600	20.6
Cholesky factorization / faer 64 × 64 · f64	14600	14.6
Cholesky factorization / nalgebra 128 × 128 · f64	121800	121.8
Cholesky factorization / NumPy 128 × 128 · f64	112900	112.9
Cholesky factorization / faer 128 × 128 · f64	59400	59.4
Cholesky factorization / Kernix 128 × 128 · f64	221900	221.9
Cholesky factorization / faer 256 × 256 · f64	311400	311.4

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky factorization / nalgebra 256 × 256 · f64	873500	873.5
Cholesky factorization / NumPy 256 × 256 · f64	988400	988.4
Cholesky factorization / Kernix 256 × 256 · f64	697000	697
Shared-variable backward / Kernix sum(X * X), 64 × 32; prepared graph · f64	9700	9.7
Shared-variable backward / Candle sum(X * X), 64 × 32; prepared graph · f64	58300	58.3
Record + backward / Kernix sum(A · W), 32 × 16 · 16 × 8 · f64	4600	4.6
Record + backward / Candle sum(A · W), 32 × 16 · 16 × 8 · f64	9900	9.9
Model inference / Candle batched normalized/sliced MLP · f64	8100	8.1
Model inference / Kernix batched normalized/sliced MLP · f64	3500	3.5
Model recording + forward / Candle batched normalized/sliced MLP · f64	10100	10.1

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Model recording + forward / Kernix batched normalized/sliced MLP · f64	4900	4.9
Model backward / Kernix prepared graph → fresh gradients · f64	6100	6.1
Model backward / Candle prepared graph → fresh gradients · f64	31900.000000000004	31.900000000000004
Parameter update / Kernix transactional SGD, fixed gradients · f64	1000	1
Coalesced tensor contraction / Kernix 16 × 32 × 8 · 32 × 8 × 16 · f64	64000	64
Coalesced tensor contraction / NumPy 16 × 32 × 8 · 32 × 8 × 16 · f64	12100	12.1
Row softmax / Candle 256 × 64 · f64	167900	167.9
Row softmax / Kernix 256 × 64 · f64	112600	112.6
Row softmax / NumPy 256 × 64 · f64	132500	132.5
Row softmax / ndarray 256 × 64 · f64	162800	162.8

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Create zeros / NumPy 64 × 64 · f32	800	0.8
Create zeros / ndarray 64 × 64 · f32	300	0.3
Create zeros / faer 64 × 64 · f32	500	0.5
Create zeros / Candle 64 × 64 · f32	500	0.5
Create zeros / Kernix 64 × 64 · f32	400	0.4
Create ones / Candle 64 × 64 · f32	700	0.7
Create ones / faer 64 × 64 · f32	500	0.5
Create ones / ndarray 64 × 64 · f32	400	0.4
Create ones / Kernix 64 × 64 · f32	500	0.5
Create ones / NumPy 64 × 64 · f32	2100	2.1